# Evaluation of Weather Sensors for Urban Environmental Monitoring at Cleveland State University


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Collaborative Research: REU Site: The Great Lakes Wind Energy Challenges

Visit our webpage for more about these instruments and the work we have completed!

## Motivation

Installing weather sensors around an urban campus is crucial for several reasons.

- They help monitor localized climate conditions which is essential for both academic research and campus safety
- Such data can offer insights into how urbanization affects local weather patterns
- This information can be used to improve energy efficiency in campus buildings and enhance the well-being of the campus community by alerting them to adverse weather conditions

## Objectives

The objectives of this project are to:

- Learn about various weather sensors, including the MetSENS500, WindSonic4, and Young anemometer
- Develop skills in programming and writing the appropriate codes to effectively collect and analyze data from these sensors
- Use the collected data to analyze different weather parameters to gain a better understanding of how weather interacts in an urban campus environment
- To determine the best locations for installing these sensors around the campus
- Install two successful stations at different points on campus
- Pave the way for three more installations within the next year
- Document findings, procedures, and codes in an organized collection

## Instruments Investigated



***Figure 1. MetSENS500***
Compact weather sensor that measures air temperature, relative humidity, barometric pressure, wind speed, wind direction, and compass readings.



***Figure 2. WindSonic4***
Two-dimensional ultrasonic anemometer designed to measure wind speed and direction.



***Figure 3. Young Ultrasonic Anemometer***
Measures three-dimensional wind speed and direction with high precision using ultrasonic technology.




***Figure 4. CS320***
Measures solar radiation using a thermopile sensor to detect shortwave radiation.



***Figure 5. HygroVUE10***
Measures air temperature and humidity.




***Figure 6. CR350***
Datalogger designed capable of measuring a wide range of environmental sensors.




***Figure 7. CR1000X***
Datalogger designed for extensive environmental monitoring and data acquisition.

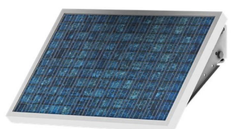


***Figure 8. SP20***
20-watt solar power solution designed to recharge batteries.

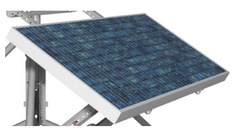


***Figure 9. SP90-L***
90-watt solar power solution designed to recharge batteries.

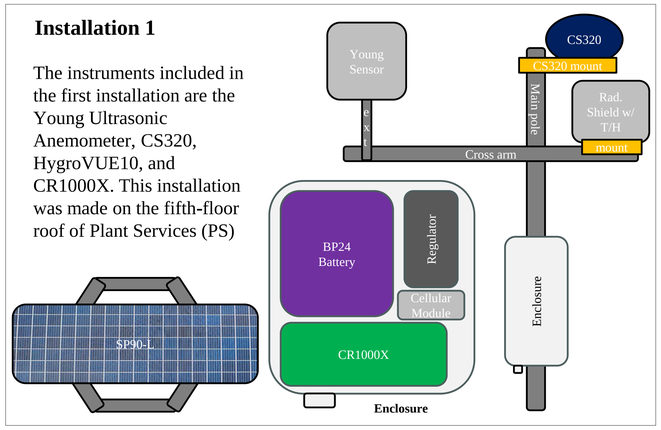
## Two Installations

**Cleveland State University Campus Map**
PS : Plant Services – Installation 1
SC : Student Center – Installation 2



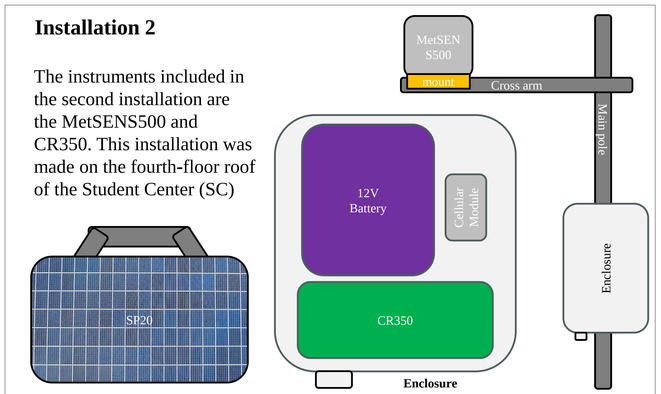

### Installation 1

The instruments included in the first installation are the Young Ultrasonic Anemometer, CS320, HygroVUE10, and CR1000X. This installation was made on the fifth-floor roof of Plant Services (PS)




### Installation 2

The instruments included in the second installation are the MetSENS500 and CR350. This installation was made on the fourth-floor roof of the Student Center (SC)




## Testing

**24BP Battery and SP90-L Configuration**

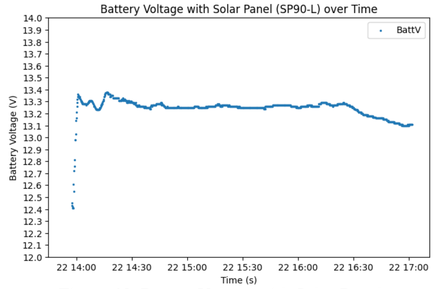



***Figure 10. Battery Voltage with Solar Panel***
Tracks battery voltage over time for a system powered by an SP90-L solar panel

**MetSENS500 vs. WindSonic4 on First Floor Roof**

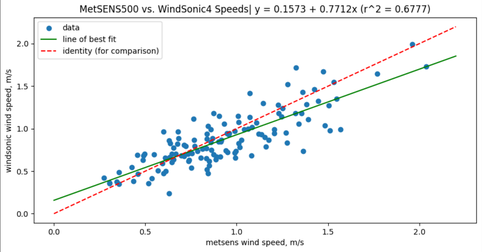



***Figure 11. MetSENS500 vs. WindSonic4***
Line of best fit indicates a strong positive correlation, suggesting both sensors are reliable for wind speed measurement

## Challenges Faced




***Figure 12. CELL205***

- CSU is not capable of supplying a static IP address to support the function of the current dataloggers in possession
- The CELL205 is the solution to the most complicated problem faced while integrating the installations
- This device makes it possible for data to be transmitted from the station to laboratory via cellular communication




***Figure 13. LoggerNet: Short Cut***

- LoggerNet comes equipped with Short Cut which can generate programs for basic SDI-12 functions
- The MetSENS500 runs on RS-232 protocol which required further intuition, language understanding, and manual references to generate a working program
- The combination of Campbell Scientific sensors with the Young Anemometer required additional manual investigation and research to run a well functioning program

## Conclusions

The evaluation of weather sensors for urban environmental monitoring at Cleveland State University assessed the performance and operations of several sensors

- The Young 81000V-90L provided precise wind data but requires further investigation with regards to its temperature measurement readings
- The WindSonic4, MetSENS500, CS320, and HygroVUE10 showed consistent values, demonstrating reliability and accuracy
- The SP20/SP90-L effectively restored energy to the rechargeable battery

Installations at the two sites on campus will will pave the way for three more installations within the next year

- Gained knowledge on how to connect sensors to dataloggers over cellular and WiFi
- Successfully integrated two protocols into one program (SDI-12 and RS-232)
- Constructed a seamless program to run the Young and Campbell Scientific instruments under the same program
- Learned to use LoggerNet, Python, GitHub, and the functionalities of the sensors as they are integrated into programs
- Created a large cache of information, procedures, findings, and resources on an online website for future student researchers in this field to continue the progress made during this term

These findings validate the effectiveness of the tested sensors for urban environmental monitoring, informing future installations at CSU and propelling the campus towards a green future


## Acknowledgements

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